

Supplementary Material

An Independent-Sample Screening–Confirmation Analysis of Casualty Occurrence in Expressway Crashes: Integrating XGBoost-SHAP with Robust Logistic Regression

Supplementary Table S1. Reproducibility specification

Item	Specification
Random seed	20260716
Split	Training 50%; validation 15%; screening 15%; confirmation 20%
Selection metric	validation PR-AUC
Threshold rule	validation-set maximum F1, then frozen
Selected model	XGB_natural
Rows / predictors / positive outcomes	83,297 / 17 / 8,993
Python	3.12.13 (main, Mar 3 2026, 15:01:35) [MSC v.1944 64 bit (AMD64)]
pandas / numpy / scipy	3.0.1 / 2.3.5 / 1.18.0
scikit-learn / XGBoost / SHAP	1.9.0 / 3.3.0 / 0.52.0
statsmodels / imbalanced-learn	0.14.6 / 0.14.2
Processed data SHA-256	c931e4c92be17c4fff3b0959794525d082569aed34a68c8da81c0d7428f2798e
Partial raw SHA-256	db671c0d5e5f5d4d9a12358ba08d9866601ba29f558e85c6911d49510c7ba37f

Note. analysis_id is deterministic for reproduction and is not the original source-system crash identifier.

Supplementary Table S2. Complete confirmation main-effect results

Contrast	n	Cases	Rate	OR	95% CI	BH q	q<0.05
Crash location: Tollgate (TG) vs Mainline	1827	51	2.79%	0.117	0.086–0.159	<0.001	Yes
Crash location: Ramp vs Mainline	2413	103	4.27%	0.304	0.245–0.377	<0.001	Yes
Primary cause: Road debris vs Speeding	1646	12	0.73%	0.071	0.039–0.128	<0.001	Yes
Traffic obstruction: Stopped vehicle vs No obstruction	314	112	35.67%	2.893	2.234–3.745	<0.001	Yes
Causal vehicle: Truck vs Passenger car	4101	617	15.05%	1.800	1.554–2.086	<0.001	Yes
Traffic obstruction: Congestion vs No obstruction	189	72	38.10%	3.170	2.300–4.370	<0.001	Yes
Primary cause: Drowsiness vs Speeding	2180	454	20.83%	1.839	1.497–2.258	<0.001	Yes
Driver condition: Other vs Normal	142	44	30.99%	3.122	2.117–4.603	<0.001	Yes
Causal vehicle: Van vs Passenger car	880	131	14.89%	1.882	1.513–2.340	<0.001	Yes
Primary cause: Single-vehicle fire vs Speeding	355	5	1.41%	0.079	0.032–0.195	<0.001	Yes
Primary cause: Inattention vs Speeding	3661	587	16.03%	1.668	1.390–2.002	<0.001	Yes
Pre-crash maneuver: Lane change vs Lane keeping	320	83	25.94%	2.029	1.527–2.695	<0.001	Yes
Causal vehicle: Other vs Passenger car	70	14	20.00%	4.086	1.973–8.464	<0.001	Yes
Traffic obstruction: Other vs No obstruction	76	25	32.89%	2.620	1.573–4.365	0.001	Yes
Primary cause: Insufficient headway vs Speeding	364	75	20.60%	1.682	1.219–2.322	0.007	Yes
Causal vehicle: Trailer vs Passenger car	936	104	11.11%	2.444	1.401–4.263	0.007	Yes
Road environment: Fallen object vs Normal	470	3	0.64%	0.157	0.050–0.499	0.007	Yes
Road environment: Pothole vs Normal	242	1	0.41%	0.043	0.006–0.307	0.007	Yes
Primary cause: Improper loading vs Speeding	197	4	2.03%	0.199	0.071–0.562	0.009	Yes
Lighting: Operating vs Not applicable (daytime)	4894	565	11.54%	1.221	1.072–1.390	0.010	Yes
Primary cause: Tire failure vs Speeding	615	51	8.29%	0.603	0.432–0.840	0.010	Yes
Primary cause: Animal intrusion vs Speeding	193	8	4.15%	0.333	0.161–0.692	0.011	Yes
Vehicle class: Large vs Medium	2135	271	12.69%	1.280	1.083–1.513	0.012	Yes
Causal vehicle: Special vehicle vs Passenger car	116	14	12.07%	2.977	1.357–6.532	0.021	Yes
Primary cause: Road condition vs Speeding	211	4	1.90%	0.248	0.086–0.716	0.030	Yes
Crash location: Service area vs Mainline	236	24	10.17%	0.562	0.361–0.874	0.031	Yes
Weather: Snow vs Clear	571	42	7.36%	0.584	0.382–0.892	0.036	Yes
Lighting: None vs Not applicable (daytime)	2452	356	14.52%	1.191	1.029–1.378	0.051	No
Vehicle class: Box truck vs Medium	225	56	24.89%	1.559	1.075–2.261	0.051	No
Primary cause: Other vehicle defect vs Speeding	101	5	4.95%	0.345	0.137–0.868	0.060	No
Pre-crash maneuver: Other vs Lane keeping	667	91	13.64%	1.340	1.036–1.734	0.064	No
Primary cause: Other vehicle factor vs Speeding	108	5	4.63%	0.350	0.137–0.894	0.067	No
Weather: Rain vs Clear	3416	251	7.35%	0.720	0.535–0.970	0.070	No
Crash location: Tunnel vs Mainline	688	103	14.97%	0.762	0.589–0.985	0.085	No
Primary cause: Unsafe passing vs Speeding	294	51	17.35%	1.449	1.015–2.068	0.089	No
Road environment: Visibility obstruction vs Normal	170	10	5.88%	0.519	0.257–1.048	0.142	No
Horizontal alignment: Unknown vs Straight	914	67	7.33%	0.733	0.517–1.039	0.162	No

Contrast	n	Cases	Rate	OR	95% CI	BH q	q<0.05
Sex: Female vs Male	2086	225	10.79%	1.163	0.982–1.377	0.162	No
Pre-crash maneuver: Excessive steering vs Lane keeping	2014	208	10.33%	0.869	0.735–1.028	0.197	No
Driver age: Under 20 vs 40s	953	76	7.97%	0.814	0.616–1.075	0.279	No
Weather: Other vs Clear	68	11	16.18%	1.720	0.794–3.726	0.313	No
Vehicle class: SUV vs Medium	1943	168	8.65%	1.143	0.942–1.385	0.317	No
Surface condition: Damp vs Dry	4292	327	7.62%	0.827	0.625–1.095	0.327	No
Primary cause: Unknown vs Speeding	98	3	3.06%	0.465	0.144–1.502	0.347	No
Horizontal alignment: Right curve vs Straight	2180	222	10.18%	0.897	0.758–1.062	0.349	No
Vehicle class: Mini vs Medium	669	67	10.01%	1.203	0.901–1.607	0.349	No
Driver age: 20s vs 40s	1690	172	10.18%	0.879	0.715–1.081	0.359	No
Road environment: Other vs Normal	244	13	5.33%	0.677	0.360–1.274	0.359	No
Embankment/cut class: Embankment <2 m vs Level	825	99	12.00%	1.151	0.908–1.457	0.379	No
Vehicle class: Unknown vs Medium	1490	148	9.93%	0.745	0.442–1.255	0.408	No
Driver condition: Fatigue vs Normal	885	197	22.26%	1.119	0.906–1.383	0.434	No
Crash location: Other vs Mainline	36	4	11.11%	0.583	0.211–1.615	0.434	No
Driver age: 50s vs 40s	3087	360	11.66%	0.918	0.781–1.080	0.434	No
Horizontal alignment: Left curve vs Straight	2119	215	10.15%	0.920	0.779–1.085	0.450	No
Day type: Weekend vs Weekday	5320	549	10.32%	0.944	0.842–1.059	0.451	No
Road environment: Slippery surface vs Normal	2283	187	8.19%	1.106	0.882–1.387	0.518	No
Embankment/cut class: Cut <10 m vs Level	1138	146	12.83%	1.091	0.894–1.332	0.518	No
Lighting: Not operating vs Not applicable (daytime)	575	51	8.87%	0.873	0.639–1.193	0.518	No
Surface condition: Snow-covered vs Dry	83	5	6.02%	0.677	0.258–1.774	0.548	No
Driver age: Unknown vs 40s	3705	324	8.74%	0.933	0.784–1.111	0.548	No
Surface condition: Other vs Dry	26	3	11.54%	1.629	0.464–5.717	0.548	No
Embankment/cut class: Embankment 2–<5 m vs Level	1942	201	10.35%	0.933	0.780–1.116	0.548	No
Pavement type: Asphalt vs Concrete	6648	719	10.82%	0.961	0.860–1.074	0.572	No
Pavement type: Unknown vs Concrete	559	43	7.69%	0.867	0.583–1.291	0.572	No
Sex: Unknown vs Male	549	40	7.29%	0.843	0.519–1.368	0.572	No
Vehicle class: Small vs Medium	3819	421	11.02%	1.051	0.909–1.214	0.575	No
Primary cause: Other vs Speeding	2238	232	10.37%	1.073	0.869–1.325	0.575	No
Driver condition: Alcohol vs Normal	420	51	12.14%	1.113	0.808–1.533	0.575	No
Embankment/cut class: Unknown vs Level	1662	150	9.03%	1.088	0.831–1.424	0.593	No
Driver age: 60 or older vs 40s	1313	176	13.40%	1.062	0.867–1.300	0.609	No
Vehicle class: General freight vs Medium	682	79	11.58%	0.923	0.692–1.229	0.623	No
Embankment/cut class: Cut ≥10 m vs Level	847	107	12.63%	1.067	0.842–1.351	0.626	No
Embankment/cut class: Embankment ≥15 m vs Level	1871	226	12.08%	0.957	0.805–1.137	0.639	No
Vehicle class: Other vs Medium	44	7	15.91%	0.903	0.346–2.355	0.857	No
Driver age: 30s vs 40s	2633	293	11.13%	0.988	0.832–1.173	0.900	No
Weather: Cloudy vs Clear	2224	249	11.20%	1.004	0.858–1.176	0.960	No

Note. HC3-robust standard errors. The table includes all estimable dummy contrasts; small cells and generic categories should be interpreted cautiously.

Supplementary Table S3. Complete SHAP screening importance

Encoded feature	Parent variable	Mean SHAP	Mean SHAP when present	Screen n
Crash location=Mainline	Crash location	0.26623	0.16920	8630
Primary cause=Road debris	Primary cause	0.25068	-1.76251	1243
Causal vehicle=Passenger car	Causal vehicle	0.17106	-0.15351	7974
Traffic obstruction=No obstruction	Traffic obstruction	0.11592	-0.07743	12076
Primary cause=Inattention	Primary cause	0.11136	0.22001	2795
Crash location=Tollgate (TG)	Crash location	0.10517	-0.69250	1337
Primary cause=Drowsiness	Primary cause	0.10475	0.37284	1628
Road environment=Normal	Road environment	0.08085	0.04272	10032
Primary cause=Single-vehicle fire	Primary cause	0.07434	-2.23419	279
Crash location=Ramp	Crash location	0.07077	-0.31099	1818
Lighting=Not applicable (daytime)	Lighting	0.06989	-0.07146	6479
Driver age=Unknown	Driver age	0.06431	-0.15746	2843
Surface condition=Dry	Surface condition	0.05980	0.04140	9183
Driver condition=Normal	Driver condition	0.04675	-0.02882	11347
Crash location=Tunnel	Crash location	0.04496	0.35568	525
Pre-crash maneuver=Excessive steering	Pre-crash maneuver	0.04116	-0.17158	1550
Primary cause=Insufficient headway	Primary cause	0.03832	0.65188	264
Road environment=Pothole	Road environment	0.02598	-1.18588	192
Primary cause=Road condition	Primary cause	0.02588	-1.30312	174
Primary cause=Speeding	Primary cause	0.02044	0.02730	3079

Encoded feature	Parent variable	Mean SHAP	Mean SHAP when present	Screen n
Primary cause=Unsafe passing	Primary cause	0.02030	0.44577	216
Causal vehicle=Van	Causal vehicle	0.01970	0.16438	660
Primary cause=Animal intrusion	Primary cause	0.01955	-1.05742	131
Weather=Snow	Weather	0.01945	-0.30382	413
Lighting=None	Lighting	0.01908	0.05185	1894
Primary cause=Tire failure	Primary cause	0.01868	-0.24997	487
Pavement type=Concrete	Pavement type	0.01854	-0.01659	6945
Pavement type=Asphalt	Pavement type	0.01831	0.02207	5101
Pre-crash maneuver=Lane change	Pre-crash maneuver	0.01719	0.35121	243
Primary cause=Improper loading	Primary cause	0.01627	-0.93624	134
Pre-crash maneuver=Lane keeping	Pre-crash maneuver	0.01590	-0.00841	10218
Driver condition=Other	Driver condition	0.01578	0.57126	118
Causal vehicle=Truck	Causal vehicle	0.01547	0.02935	3034
Road environment=Fallen object	Road environment	0.01490	-0.41742	350
Weather=Rain	Weather	0.01461	-0.03962	2598
Road environment=Slippery surface	Road environment	0.01448	0.02483	1640
Crash location=Service area	Crash location	0.01353	0.33027	170
Driver age=50s	Driver age	0.01279	0.03403	2218
Sex=Unknown	Sex	0.01250	-0.21103	465
Driver age=Under 20	Driver age	0.01247	-0.12029	704
Primary cause=Other	Primary cause	0.01198	0.03019	1671
Driver age=30s	Driver age	0.01152	-0.02984	2062
Vehicle class=General freight	Vehicle class	0.01151	-0.12273	500
Vehicle class=Medium	Vehicle class	0.01149	0.01620	4228
Weather=Cloudy	Weather	0.01071	0.04003	1686
Surface condition=Damp	Surface condition	0.01066	-0.02326	3213
Vehicle class=Large	Vehicle class	0.01015	0.02131	1617
Driver age=60 or older	Driver age	0.00952	0.05054	975
Embankment/cut class=Level	Embankment/cut class	0.00946	0.00369	6297
Horizontal alignment=Right curve	Horizontal alignment	0.00930	0.03538	1608
Sex=Female	Sex	0.00877	0.03629	1595
Primary cause=Pothole	Primary cause	0.00876	-0.58784	148
Vehicle class=Small	Vehicle class	0.00842	0.00088	2867
Day type=Weekend	Day type	0.00812	0.01014	3984
Weather=Clear	Weather	0.00765	0.00475	7744
Primary cause=Other vehicle factor	Primary cause	0.00712	-0.58459	87
Driver age=40s	Driver age	0.00684	0.01133	2468
Horizontal alignment=Left curve	Horizontal alignment	0.00682	0.01915	1650
Driver age=20s	Driver age	0.00667	0.03027	1225
Causal vehicle=Trailer	Causal vehicle	0.00665	0.04127	681
Vehicle class=Unknown	Vehicle class	0.00631	-0.00104	1137
Embankment/cut class=Cut ≥10 m	Embankment/cut class	0.00611	0.05403	647
Lighting=Operating	Lighting	0.00600	0.00774	3682
Embankment/cut class=Embankment ≥15 m	Embankment/cut class	0.00593	0.02274	1388
Horizontal alignment=Straight	Horizontal alignment	0.00573	-0.00078	8604
Vehicle class=SUV	Vehicle class	0.00529	0.02234	1443
Embankment/cut class=Unknown	Embankment/cut class	0.00509	-0.02689	1198
Traffic obstruction=Stopped vehicle	Traffic obstruction	0.00454	0.08444	233
Embankment/cut class=Embankment 2-<5 m	Embankment/cut class	0.00334	-0.01223	1456
Pre-crash maneuver=Other	Pre-crash maneuver	0.00313	0.03330	484
Primary cause=Unknown	Primary cause	0.00312	-0.32972	78
Embankment/cut class=Cut <10 m	Embankment/cut class	0.00252	0.01730	849
Primary cause=Other vehicle defect	Primary cause	0.00241	-0.19199	81
Road environment=Visibility obstruction	Road environment	0.00212	0.02789	114
Sex=Male	Sex	0.00193	0.00071	10435
Driver condition=Fatigue	Driver condition	0.00160	0.01185	689
Vehicle class=Box truck	Vehicle class	0.00157	0.04737	152
Lighting=Not operating	Lighting	0.00152	-0.02039	440
Traffic obstruction=Congestion	Traffic obstruction	0.00143	0.03870	127
Weather=Other	Weather	0.00142	0.13427	54
Pavement type=Unknown	Pavement type	0.00130	0.01787	449
Embankment/cut class=Embankment <2 m	Embankment/cut class	0.00128	-0.00996	660
Day type=Weekday	Day type	0.00124	-0.00015	8511
Vehicle class=Mini	Vehicle class	0.00114	0.00861	529
Driver condition=Alcohol	Driver condition	0.00103	-0.00432	341

Encoded feature	Parent variable	Mean SHAP	Mean SHAP when present	Screen n
Causal vehicle=Other	Causal vehicle	0.00100	0.09836	60
Causal vehicle=Special vehicle	Causal vehicle	0.00090	-0.06557	86
Road environment=Other	Road environment	0.00053	-0.01481	167
Traffic obstruction=Other	Traffic obstruction	0.00040	-0.01916	59
Horizontal alignment=Unknown	Horizontal alignment	0.00036	-0.00298	633
Surface condition=Snow-covered	Surface condition	0.00016	0.00970	78
Crash location=Other	Crash location	0.00000	nan	15
Surface condition=Other	Surface condition	0.00000	nan	21
Vehicle class=Other	Vehicle class	0.00000	nan	22

Note. SHAP values were computed only in the independent screening set and are model contributions, not causal effects.

Supplementary Table S4. All screened interaction candidates

Candidate	Screen n	Confirm n	Cases	Joint rate	OR ratio	95% CI	BH q	Decision
Causal vehicle=Van × Vehicle class=Large	30	187	54	28.88%	3.007	1.877–4.815	<0.001	Confirmed
Primary cause=Inattention × Crash location=Tollgate (TG)	139	866	21	2.42%	0.411	0.228–0.741	0.008	Confirmed
Crash location=Tollgate (TG) × Vehicle class=Small	33	268	14	5.22%	2.638	1.362–5.112	0.008	Confirmed
Primary cause=Inattention × Crash location=Ramp	49	366	20	5.46%	0.787	0.471–1.314	0.539	Not confirmed
Primary cause=Inattention × Pre-crash maneuver=Excessive steering	76	477	69	14.47%	1.096	0.771–1.559	0.731	Not confirmed
Primary cause=Drowsiness × Lighting=None	51	423	113	26.71%	0.969	0.721–1.301	0.833	Not confirmed
Primary cause=Road debris × Sex=Female	30	173	2	1.16%	—	—	—	Excluded by cell rule
Primary cause=Road debris × Horizontal alignment=Left curve	28	189	2	1.06%	—	—	—	Excluded by cell rule
Primary cause=Road debris × Embankment/cut class=Embankment ≥15 m	26	189	1	0.53%	—	—	—	Excluded by cell rule
Crash location=Ramp × Vehicle class=Large	28	186	7	3.76%	—	—	—	Excluded by cell rule
Primary cause=Road debris × Pavement type=Asphalt	85	547	5	0.91%	—	—	—	Excluded by cell rule
Primary cause=Road debris × Lighting=Operating	52	395	1	0.25%	—	—	—	Excluded by cell rule
Crash location=Tollgate (TG) × Causal vehicle=Trailer	46	346	4	1.16%	—	—	—	Excluded by cell rule

Note. Confirmation testing required joint $n \geq 100$, events ≥ 10 , and non-events ≥ 10 . BH-FDR was applied to all six candidates that satisfied these rules.

Supplementary Table S5. Complete separation audit

Dummy	n	Cases	Reason
Primary cause=Pothole	194	0	complete separation or absent category in confirmation set

Note. Separated or absent contrasts were removed before estimation and were not treated as confirmed effects.

Supplementary Figures

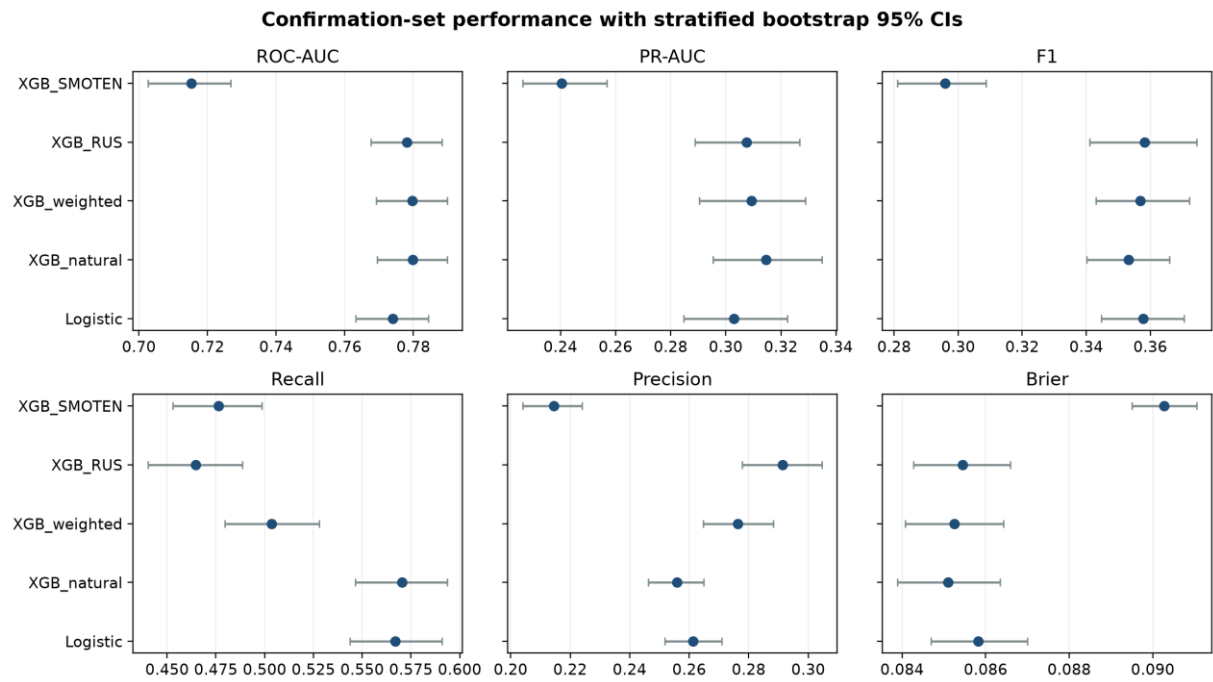


Figure S1. Confirmation-set model metrics with 95% bootstrap confidence intervals.